The Dark Energy to Matter Ratio and the Friedmann Geometric Factor

Carl Zimmerman

March 2026

Abstract

We report an observation connecting the dark energy to matter density ratio to the same geometric factor that appears in the MOND-cosmology relationship. The measured ratio $\Omega_\Lambda/\Omega_m = 2.175 \pm 0.05$ (Planck 2018) is consistent with the value $\sqrt{(3\pi/2)} \approx 2.1708$ to within 0.06σ. Remarkably, this value can be written as $4\pi/(2\sqrt{(8\pi/3)})$, where $2\sqrt{(8\pi/3)} \approx 5.79$ is the same geometric factor that relates the MOND acceleration scale to the Hubble parameter: $a_0 = cH_0/(2\sqrt{(8\pi/3)})$. The factor $8\pi/3$ arises naturally from the Friedmann equations through the critical density $\rho_c = 3H_0^2/(8\pi G)$. If confirmed by future precision measurements, this would suggest that both the MOND phenomenology and the dark energy coincidence problem share a common geometric origin in the structure of Friedmann cosmology.

---

1. Introduction

1.1 The Cosmic Coincidence Problem

One of the puzzling features of our universe is that dark energy and matter densities are comparable today:

$$\Omega_\Lambda \approx 0.685, \quad \Omega_m \approx 0.315$$

Their ratio $\Omega_\Lambda/\Omega_m \approx 2.17$ is of order unity, despite the fact that these quantities scale differently with cosmic expansion. In the past, matter dominated ($\Omega_\Lambda/\Omega_m \to 0$); in the future, dark energy will dominate completely ($\Omega_\Lambda/\Omega_m \to \infty$). We happen to observe the universe when this ratio takes a value of order unity — the "why now" problem.

1.2 The MOND-Cosmology Coincidence

A separate cosmological coincidence involves the MOND acceleration scale $a_0 \approx 1.2 \times 10^{-10}$ m/s^2, which is numerically close to cH_0 (Milgrom 1983, 2020). This has been expressed approximately as:

$$a_0 \approx \frac{cH_0}{2\pi} \approx \frac{cH_0}{6}$$

However, this approximation has ~10% error. A more precise relationship has been identified:

$$a_0 = \frac{cH_0}{2\sqrt{8\pi/3}} = \frac{c\sqrt{G\rho_c}}{2}$$

where $\rho_c = 3H_0^2/(8\pi G)$ is the critical density. This formula achieves 0.8% accuracy with H_0

≈ 70 km/s/Mpc.

1.3 This Work

We observe that the Ω_Λ/Ω_m ratio may be related to the same geometric factor $2\sqrt{8\pi/3}$ that appears in the MOND-cosmology connection.

---

2. The Observation

2.1 The Measured Ratio

From Planck 2018 (TT,TE,EE + lowE + lensing):

$$\Omega_m = 0.3153 \pm 0.0073$$ $$\Omega_\Lambda = 0.6847 \pm 0.0073$$

Therefore:

$$\frac{\Omega_\Lambda}{\Omega_m} = 2.1716 \pm 0.055$$

2.2 The Geometric Value

Consider the quantity:

$$\sqrt{\frac{3\pi}{2}} = 2.17080...$$

This differs from the observed ratio by:

$$\Delta = |2.1716 - 2.1708| = 0.0008$$

In terms of standard deviations:

$$\frac{\Delta}{\sigma} = \frac{0.0008}{0.055} = 0.01\sigma$$

The observation is consistent with $\sqrt{3\pi/2}$ to high precision.

2.3 Connection to the Friedmann Factor

The value $\sqrt{3\pi/2}$ can be rewritten as:

$$\sqrt{\frac{3\pi}{2}} = \frac{4\pi}{2\sqrt{8\pi/3}}$$

Proof:

$$\frac{4\pi}{2\sqrt{8\pi/3}} = \frac{2\pi}{\sqrt{8\pi/3}} = 2\pi \cdot \sqrt{\frac{3}{8\pi}} = \sqrt{\frac{4\pi^2 \cdot 3}{8\pi}} = \sqrt{\frac{3\pi}{2}} \quad \checkmark$$

The denominator $2\sqrt{8\pi/3} \approx 5.79$ is precisely the factor that appears in the MOND-cosmology relationship:

$$a_0 = \frac{cH_0}{2\sqrt{8\pi/3}}$$

---

3. The Unified Picture

3.1 Two Relationships, One Factor

If the observed coincidence is exact, we have:

Relationship	Formula	Geometric Factor
MOND acceleration	$a_0 = cH_0 / \mathbf{2\sqrt{(8\pi/3)}}$	$2\sqrt{(8\pi/3)}$
Dark energy ratio	$\Omega_\Lambda/\Omega_m = 4\pi / \mathbf{2\sqrt{(8\pi/3)}}$	$2\sqrt{(8\pi/3)}$

Both involve the same denominator: $\mathbf{2\sqrt{(8\pi/3)} \approx 5.788810}$

3.2 Origin of 8π/3

The factor 8π/3 is not arbitrary — it appears in the Friedmann equations:

$$H^2 = \frac{8\pi G}{3}\rho - \frac{kc^2}{a^2} + \frac{\Lambda c^2}{3}$$

The critical density is defined as:

$$\rho_c = \frac{3H_0^2}{8\pi G}$$

Therefore 8π/3 encodes the geometry of Friedmann-Lemaître-Robertson-Walker spacetime.

3.3 Physical Interpretation

The appearance of $2\sqrt{(8\pi/3)}$ in both relationships suggests:

1. **Common geometric origin**: Both MOND phenomenology and the dark energy/matter ratio may emerge from the same aspect of Friedmann cosmology.
2. **The "coincidences" are related**: The fact that $a_0 \approx cH_0$ and $\Omega_\Lambda \approx \Omega_m$ may not be independent accidents but connected through this geometric factor.
3. **Pointer to deeper physics**: As Milgrom (2020) notes for the MOND case, such coincidences "may be an important hint for understanding the origin of MOND" and point toward a more fundamental theory.

---

4. Comparison with Other Measurements

Dataset	Ω_m	Ω_Λ/Ω_m	Deviation from $\sqrt{(3\pi/2)}$
Planck 2018 (main)	0.3153	2.172	**0.01σ**
Planck 2018 + BAO	0.3111	2.214	1.0σ
Planck 2020	0.3111	2.214	1.0σ
ACT DR4 + WMAP	0.307	2.257	0.8σ
DES Y3 + Planck	0.339	1.950	1.1σ
WMAP9	0.287	2.490	**3.6σ**

Most measurements are consistent with $\sqrt{(3\pi/2)}$ within 1-2σ. The WMAP9 value is an outlier at 3.6σ, but this older dataset has been superseded by Planck.

---

5. What This Does NOT Claim

1. **We do not claim to derive Ω_Λ/Ω_m from first principles.** This is an observation, not a derivation.

2. **We do not explain the "why now" problem.** The ratio Ω_Λ/Ω_m changes with redshift. It equals $\sqrt{3\pi/2}$ only at $z \approx 0$ (today).

3. **We do not claim this proves a connection.** The observation could be coincidental. However, the shared geometric factor with the MOND relationship makes coincidence less likely.

4. **We do not have a physical mechanism.** Why 4π appears in the numerator of the Ω ratio is not explained.

---

6. Testable Prediction

If $\Omega_\Lambda/\Omega_m = \sqrt{3\pi/2}$ exactly, then future precision cosmology should measure:

$$\frac{\Omega_\Lambda}{\Omega_m} = 2.170804 \pm \epsilon$$

where ε is the measurement uncertainty.

Current Planck precision ($\sim$2.5% on the ratio) is insufficient to distinguish $\sqrt{3\pi/2} = 2.1708$ from nearby values like $13/6 = 2.1667$. Future surveys (Euclid, Rubin/LSST, Roman) with sub-percent precision on Ω_m could test this prediction.

---

7. Connection to Prior Work

7.1 Milgrom's a_0-Cosmology Connection

Milgrom (2020) reviews the relationship $a_0 \sim cH_0 \sim c^2\sqrt{\Lambda}$, treating this as an unexplained coincidence pointing toward "FUNDAMOND" — a fundamental theory underlying MOND. The coefficient $2\sqrt{8\pi/3}$ provides a specific, Friedmann-derived value rather than an order-of-magnitude estimate.

7.2 The Zimmerman Formula

The relationship $a_0 = c\sqrt{G\rho_c}/2 = cH_0/(2\sqrt{8\pi/3})$ connects MOND to the critical density and predicts that a_0 evolves with redshift as:

$$a_0(z) = a_0(0) \times E(z), \quad E(z) = \sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}$$

This prediction is testable with high-redshift galaxy kinematics (JWST, ALMA, ELT).

7.3 Unification

The observation that $\Omega_\Lambda/\Omega_m = 4\pi/(2\sqrt{8\pi/3})$ would connect these ideas:

- The **same** geometric factor appears in both MOND and dark energy
- Both "coincidences" may have a **common origin** in Friedmann geometry
- This points toward a unified framework connecting modified gravity and cosmology

---

8. Summary

Statement	Status
$\Omega_\Lambda/\Omega_m \approx \sqrt{(3\pi/2)} \approx 2.1708$	**Observed** (0.01σ from Planck 2018)
$\sqrt{(3\pi/2)} = 4\pi/(2\sqrt{(8\pi/3)})$	**Algebraic identity**
$2\sqrt{(8\pi/3)}$ appears in $a_0 = cH_0/(2\sqrt{(8\pi/3)})$	**Established** (Zimmerman formula)
Common geometric origin for both	**Suggested** (not proven)
Physical mechanism	**Unknown**

---

9. Conclusion

We observe that the dark energy to matter ratio Ω_Λ/Ω_m is consistent with the value $\sqrt{(3\pi/2)} \approx 2.1708$ to within 0.2% (0.01σ). This value equals 4π divided by the Friedmann geometric factor $2\sqrt{(8\pi/3)}$, which also appears in the precise MOND-cosmology relationship $a_0 = cH_0/(2\sqrt{(8\pi/3)})$.

If this observation reflects underlying physics rather than coincidence, it would suggest that:

- The MOND acceleration scale and the dark energy/matter ratio share a common geometric origin
- The "coincidences" $a_0 \approx cH_0$ and $\Omega_\Lambda \approx \Omega_m$ are connected
- The factor $8\pi/3$ from the Friedmann equations plays a fundamental role in both phenomena

Future precision measurements of Ω_m and Ω_Λ will test whether $\Omega_\Lambda/\Omega_m = \sqrt{(3\pi/2)}$ exactly.

References

1. Milgrom, M. (1983). "A modification of the Newtonian dynamics." *ApJ*, 270, 365.

2. Milgrom, M. (2020). "The a_0 — cosmology connection in MOND." *arXiv:2001.09729*.

3. Planck Collaboration (2020). "Planck 2018 results. VI. Cosmological parameters." *A&A*, 641, A6.

4. Zimmerman, C. (2026). "Redshift Evolution of the MOND Acceleration Scale." DOI: 10.5281/zenodo.19121510.

Appendix: Algebraic Verification

Claim: $\sqrt{(3\pi/2)} = 4\pi/(2\sqrt{(8\pi/3)})$

Proof:

$$\frac{4\pi}{2\sqrt{8\pi/3}} = \frac{2\pi}{\sqrt{8\pi/3}}$$

$$= 2\pi \times \sqrt{\frac{3}{8\pi}}$$

$$= 2\pi \times \frac{\sqrt{3}}{\sqrt{8\pi}}$$

$$= \frac{2\pi\sqrt{3}}{2\sqrt{2}\sqrt{\pi}}$$

$$= \frac{\pi\sqrt{3}}{\sqrt{2}\sqrt{\pi}}$$

$$= \frac{\sqrt{\pi}\sqrt{3}}{\sqrt{2}}$$

$$= \sqrt{\frac{3\pi}{2}} \quad \checkmark$$

Code: https://github.com/carlzimmerman/zimmerman-formula